No.	Status	Date
LNDE-RPT-20251103-041-DS-2-RF-102	Issued	2025-11-03

UT Thickness Data Sheet - 2-RF-102

Scope and Responsibilities

This data sheet provides the ultrasonic thickness measurement results for the reflux line 2-RF-102 located within the ATU-100. The scope of work consists of a 12-month re-measurement of established condition monitoring locations to track wall loss and determine remaining life.

Sarah Thompson, acting as the UT technician (SNT-TC-1A UT Level II) for Larkhall NDE Services Inc., performed the measurements and data collection. All activities were conducted in accordance with the requirements of QSCP-NDT-001 Rev 4 and ASTM E797.

Technical Specifications

The subject line 2-RF-102 is constructed from A312 TP316 with a nominal wall thickness of 3.91 mm and an outside diameter of 60.3 mm. The design limits for this service are 517 kPa(g) and 104 °C.

The minimum allowable thickness (t-min) is 2.16 mm. Measurements were performed using the stainless-steel velocity setting to ensure accuracy for the A312 TP316 material.

Measurement Results

Thickness readings were recorded at four specific CMLs. The measured values are 1.6 mm at 2-RF-102-04, 1.69 mm at 2-RF-102-05, 1.67 mm at 2-RF-102-06, and 1.75 mm at 2-RF-102-07.

All four recorded thicknesses are below the t-min of 2.16 mm. Consequently, the remaining life for these locations is expired.

Analysis and Evaluation

As detailed in LNDE-RPT-20251103-041-R0, the corrosion rate remains unchanged. Despite the readings being below the structural limit, the projected turnaround date confirms that the replacement schedule remains unchanged.

The current wall thicknesses at 2-RF-102-04 through 2-RF-102-07 are below the retirement thickness, necessitating the issuance of an NCR per site rules.

Thickness readings, measured 2025-11-03

Line 2-RF-102 | Reflux | A312 TP316 | t_nom 3.91 mm | t-min 2.16 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RF-102-01	3.91	2.16	2024-11-04	3.55	3.52	0.013	0.023	>50	2031-11-03	
2-RF-102-02	3.91	2.16	2024-11-04	3.53	3.49	0.013	0.026	>50	2031-11-03	
2-RF-102-03	3.91	2.16	2024-11-04	3.65	3.64	0.010	0.007	>50	2031-11-03	
2-RF-102-04	3.91	2.16	2024-11-04	1.94	1.60	0.077	0.203	expired	2026-11-03	< t-min; gauge photo attached
2-RF-102-05	3.91	2.16	2024-11-04	1.92	1.69	0.165	0.197	expired	2026-11-03	< t-min; gauge photo attached
2-RF-102-06	3.91	2.16	2024-11-04	1.92	1.67	0.168	0.177	expired	2026-11-03	< t-min; gauge photo attached
2-RF-102-07	3.91	2.16	2024-11-04	1.86	1.75	0.075	0.177	expired	2026-11-03	< t-min; gauge photo attached

Sign-off	Name	Function	Date	Signature
Prepared	Sarah Thompson	UT technician, SNT-TC-1A UT Level II	2025-11-03	_____